

UNIT 6 • GUIDED NOTES

Unit 6 — Functions & *Linear Relationships*

Interactive guided notes for Unit 6 — real-world context, comparing functions, and qualitative graphs. Fill in the blanks to learn — your progress saves automatically.

📁 Unit 6 — All Topics

Your complete Unit 6 guided notes. Topics 6.4–6.6 are interactive on this page; 6.1, 6.2, 6.3, and 6.7 open their full guided-notes pages.

6.1 Functions — relations, mappings, function rule

OPEN NOTES →

6.2 Slope — rise/run, the slope formula

OPEN NOTES →

6.3 Tables → Equations — write a linear rule from a table

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6.4 Linear Functions in Real-World Context

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6.5 Comparing Functions

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6.6 Qualitative Graphs

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6.7 Systems of Equations — graphing & substitution

OPEN NOTES →

🎮 Games & Activities

Self-paced games and activities that reinforce Unit 6 — a class favorite.

🔍 Line Detective Activity

→

🚀 UFO Blaster Game

→

📈 Slope-Intercept Intro

→

🗨️ Slope-Intercept Practice

→

📊 Tables & Linear Functions

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★ Unit 6 Interactive Review (25 Q)

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6.4

Linear Functions in Real-World Context

NC.8.F.4 — CONSTRUCT A FUNCTION TO MODEL A LINEAR RELATIONSHIP; INTERPRET RATE OF CHANGE AND INITIAL VALUE IN CONTEXT.

What is this section about? Real-world problems often follow a linear pattern — a steady rate of change plus a starting value. We can write equations in $y = mx + b$ form to model these situations and use them to make predictions.

Slope & Y-Intercept in Context

In real-world problems, the slope and y-intercept have specific meanings:

m (slope) = the **show** of change — how much y changes per unit of x.

b (y-intercept) = the **show** value — what y equals when x = **show** .

 **Quick check:** The y-intercept is always the "before-anything-happens" value — the initial cost, starting amount, or fixed fee.

Keywords for Slope and Y-Intercept

Slope keywords (rate):

- per hour, per mile, per day
- each, every
- increases by, decreases by
- price/cost per item

Y-intercept keywords (start):

- flat fee, base cost
- starting amount, initial
- up-front, deposit
- "to begin with"

EXAMPLE

Plumber example: A plumber charges \$50 to come to your house, plus \$30 per hour of work.

The flat fee (\$50) is paid before any work happens → $b = 50$

The \$30 per hour is the rate of change → $m = 30$

Equation: $y = 30x + 50$

Interpret: If $x = 4$ hours, then $y = 30(4) + 50 = \$170$ total.

TRY IT

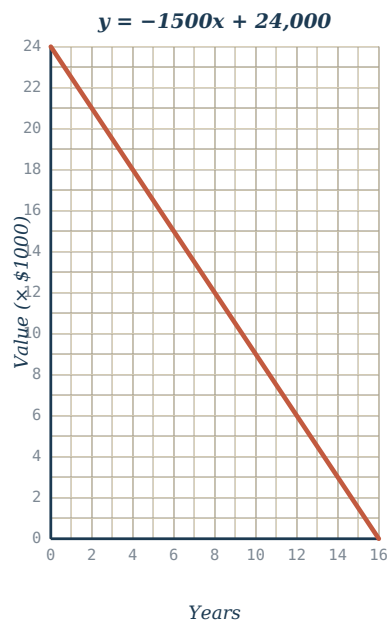
Practice 1: A gym charges a \$40 sign-up fee plus \$25 per month. Write the equation in slope-intercept form for the total cost y after x months.

Slope (m) = show · Y-intercept (b) = show

Equation: $y =$ $x +$

EXAMPLE

Car value example: The value of a car after x years is modeled by $y = -1500x + 24,000$.



$m = -1500 \rightarrow$ the car loses **\$1,500 per year** (slope is negative because value decreases)
 $b = 24,000 \rightarrow$ the car was worth **\$24,000** when new (when $x = 0$)

TRY IT

Practice 2: A swimming pool is being filled. After 3 minutes there are 90 gallons; after 8 minutes there are 240 gallons. The pool was empty before filling started.

Find the slope (gallons per minute): $m =$ show

The y-intercept (initial gallons) = show

Equation: $y =$ x

Hint: slope = $(240 - 90) \div (8 - 3) = 150 \div 5 = ?$

⚠ Watch the sign of the slope: If the quantity is *increasing* over time (filling, growing, earning), m is **positive**. If *decreasing* (depreciating, draining, using up), m is **negative**.

TRY IT

Practice 3: Your phone has 80% battery and loses 5% per hour of use. Write an equation for battery percent y after x hours.

$m =$ · $b =$

Equation: $y =$ $x +$

After 4 hours, battery = %

6.5

Comparing Functions

NC.8.F.2 – COMPARE PROPERTIES OF TWO FUNCTIONS EACH REPRESENTED IN A DIFFERENT WAY (ALGEBRAICALLY, GRAPHICALLY, IN A TABLE, OR VERBALLY).

The big idea: Linear functions can be shown four ways — equation, table, graph, or verbal description. To compare two functions, get both into $y = mx + b$ form (or just identify their slopes and y-intercepts). Then you can compare directly.

The Strategy

Before answering *any* comparison question, find the slope (m) and y-intercept (b) for **BOTH** functions first.

To compare *rate of change*, look at:

To compare *starting value*, look at:

The function with the steeper graph has the slope.

EXAMPLE

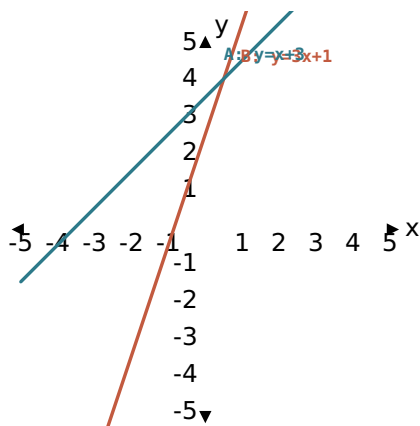
Compare these two functions:

Function A (equation):

$$y = x + 3$$

Function B (table):

x	0	1	2	3
y	1	4	7	10



Function A: $m = 1$, $b = 3$

Function B: $m = 3$ (each row, y goes up by 3); $b = 1$ (when $x = 0$)

Greater rate of change? Function B (slope of 3 > slope of 1).

Higher starting value? Function A ($b = 3 > b = 1$).

TRY IT

Practice 1: Compare these two functions:

Function A:

$$y = 4x + 2$$

Function B:

x	0	1	2	3
y	5	7	9	11

Function A: $m =$, $b =$

Function B: $m =$, $b =$

Which has the higher rate of change? Function [show](#)

Which has the higher starting value? Function [show](#)

EXAMPLE

Cell phone plans: Plan A costs \$20/month plus \$0.05/minute. Plan B costs \$30/month plus \$0.03/minute.

Plan A: $y = 0.05x + 20$

Plan B: $y = 0.03x + 30$

At $x = 0$ minutes: Plan A = \$20, Plan B = \$30 → Plan A starts cheaper

At $x = 500$ minutes: Plan A = \$45, Plan B = \$45 → tied!

At $x = 1000$ minutes: Plan A = \$70, Plan B = \$60 → Plan B is cheaper for heavy use

TRY IT

Practice 2: Two savings accounts:

- Account A starts with \$300 and adds \$25/week → $y = 25x + 300$
- Account B starts with \$400 and adds \$20/week → $y = 20x + 400$

Which has more money after 10 weeks?

Account A after 10 weeks: $y = 25($ $) + 300 = \$$ [show](#)

Account B after 10 weeks: $y = 20(10) + 400 = \$$ [show](#)

Account [show](#) has more after 10 weeks.

• **Strategy:** When word problems compare two situations, write *both* equations first, then plug in the same value of x to compare. Or, if asked about long-term, look at the slopes — steeper slope wins eventually.

6.6

Qualitative Graphs

NC.8.F.5 — DESCRIBE QUALITATIVELY THE FUNCTIONAL RELATIONSHIP BETWEEN TWO QUANTITIES BY ANALYZING A GRAPH; SKETCH A GRAPH FROM A VERBAL DESCRIPTION.

What's a qualitative graph? A graph that shows the *shape* of a relationship over time without exact numbers. The story of the graph matters more than specific values. Common examples: distance-time graphs, temperature-time graphs.

Reading the Shape

Each section of a qualitative graph tells you something. Going left-to-right, ask:

Going UP means the quantity is [show](#) .

Going DOWN means the quantity is [show](#) .

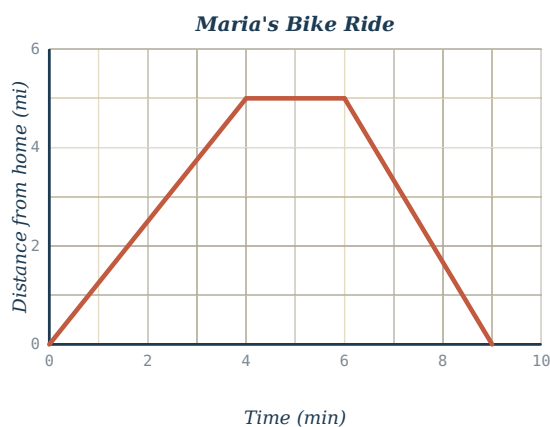
FLAT (horizontal) means the quantity is [show](#) (not changing).

STEEPER lines mean the change is [show](#) .

SHALLOWER (gentler) lines mean the change is [show](#) .

EXAMPLE

Maria rides her bike to school:



The graph shows Maria's distance from home over time. We can read the story:

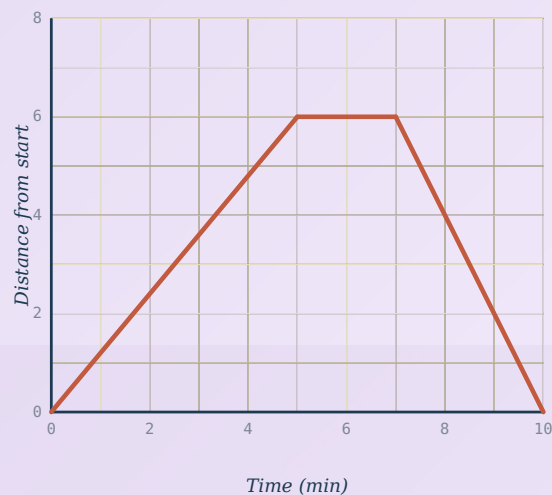
- **0 to 4 min:** line goes UP at a steady rate → Maria is biking away from home toward school.
- **4 to 6 min:** line is FLAT → Maria stays at school (distance doesn't change).
- **6 to 9 min:** line goes DOWN steeper than it went up → Maria rides home *faster* than she went.

⚠ **Common mistake:** When someone is *returning home*, the distance-from-home line goes DOWN (toward zero), not back up! The y-axis is "distance from start" — if you're getting closer to home, your distance is decreasing.

TRY IT

Practice 1: A swimmer dives into a pool, swims at a steady pace to the far wall, rests for a moment, then swims back *faster* than she went.

The graph below should show distance from start versus time. Match each phase to the correct description:



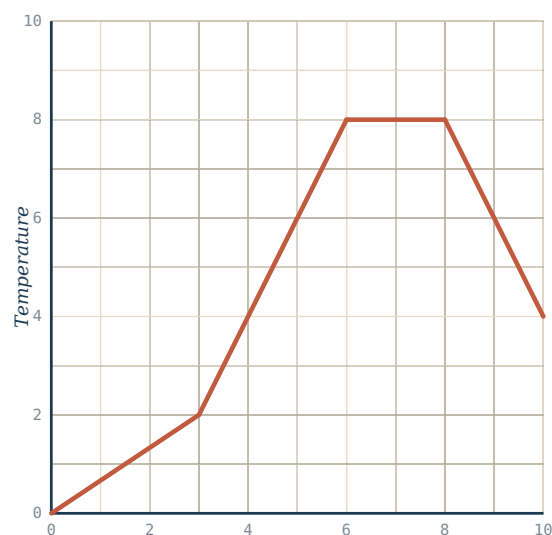
Phase 1 (going up at steady rate): she swam [show](#) to the far wall at a steady pace.

Phase 2 (flat line): she [show](#) at the far wall.

Phase 3 (going down steeper): she swam [show](#) faster than she went.

EXAMPLE

A pot of water: A pot of water is placed on a stove. It heats slowly at first, then more rapidly, stays at boiling for several minutes, then is removed and cools gradually.



Time (min)

- **Phase 1 (shallow up):** heats slowly
- **Phase 2 (steep up):** heats more rapidly
- **Phase 3 (flat):** at boiling, temperature stays constant
- **Phase 4 (down):** removed from stove, cools off gradually

TRY IT

Practice 2: Look at a distance-time graph with three phases:

- Phase 1 (0–3 min): steep upward line
- Phase 2 (3–6 min): flat line
- Phase 3 (6–10 min): gradual upward line

Phase 1: the person is moving [show](#) (away from start).

Phase 2: the person is [show](#) (not moving).

Phase 3: the person is moving [show](#) (still away from start).

📐 **Steps for sketching from a story:**

1. Break the story into 2–4 phases.
2. For each phase, decide: increase, decrease, or stay the same?
3. Decide if the change is fast (steep) or slow (shallow).
4. Connect phases at their endpoints — no jumps in the line!